

Pre-registration — DentaCoPilot

Clinician-in-the-loop evaluation, BigMouth multi-institutional cohort, KLE V K Institute of Dental Sciences clinician arm

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Status: *Draft. The final version of this document will be deposited on the Open Science Framework (OSF) before any clinician participates and before the test split of the BigMouth or KLE data is read. The OSF DOI will be cited in the final manuscript’s methods section.*

1 Research questions

- **RQ1.** Does an LLM-based next-procedure recommender (DentaCoPilot, arm C) improve dentists’ top-1 agreement with retrospective ground-truth treatment relative to dentist-alone (arm A) and a classical XGBoost recommender (arm B)?
- **RQ2.** Does it reduce time-to-decision relative to arms A and B?
- **RQ3.** Is calibrated abstention (arm C abstains when confidence below τ) associated with appropriate dentist behavior — i.e., do dentists override less often on covered cases than on cases where the system abstains?

2 Hypotheses

H1 (primary). Mean top-1 agreement is highest in arm C, with the contrast C vs A statistically significant at $\alpha = 0.05$, two-sided.

H2 (secondary). Mean time-to-decision in arm C is at most 10% higher than in arm A. We pre-commit to a non-inferiority contrast on time, so we cannot win on accuracy by burying a time regression.

H3 (exploratory). Override rate on covered cases is below override rate on abstained cases.

3 Design

Within-subject crossover. Each dentist sees three blocks of ~ 33 cases under arms A / B / C, block-to-arm randomized per dentist via Latin square. Cases are stratified to span all nine CDT categories with weights matching KLE’s actual case mix.

4 Sample size

Target $N = 12$ dentists $\times \sim 33$ cases per arm ≈ 400 case-arm observations per arm. Mixed-effects logistic regression (random intercept per dentist) gives $\geq 80\%$ power to detect a Cohen’s $d = 0.4$ difference in agreement at $\alpha = 0.05$. If recruitment falls below $N = 8$, the study is reframed as a *pilot* with descriptive results only and **no inferential claims**.

5 Primary outcome

Top-1 agreement (binary) between dentist’s pick and retrospective ground-truth procedure code (exact CDT match), per case-arm observation. Analyzed via mixed-effects logistic regression with random intercepts for dentist and for case, fixed effect for arm (treatment-coded with arm A as reference).

6 Secondary outcomes

- S1.** Top-1 agreement allowing CDT *category* match instead of exact code (more lenient definition).
- S2.** Time-to-decision (seconds), log-transformed for analysis. Random intercept per dentist.
- S3.** Override rate in arms B and C: probability that the dentist’s final pick differs from the recommender’s top-1, given that the recommender provided a recommendation.
- S4.** Trust (Likert 1–5), per case, per arm where applicable.

7 Models trained on the dev split (locked specification)

- B0 — frequency baseline.
- B1 — TF-IDF + multinomial logistic regression on chart-token features.
- B2 — XGBoost on engineered chart-history features.
- B3 — MultiTP-style CNN–RNN reimplementation (Liu et al. AIM 2024).
- M1 — Claude Haiku, structured prompt, top-K + verbalised confidence + abstain.
- M2 — Claude Sonnet + chain-of-thought.
- M3 — Claude Sonnet + retrieval over the open clinical evidence corpus.
- M4 — Claude Opus + chain-of-thought.
- M5 — Claude Sonnet + chain-of-thought + **classical prior** (proposed forward design).
- M6 — Claude Opus + chain-of-thought + classical prior.

For the clinical-evaluation arm (B vs C contrast), the *frozen* models are: **B = B2 (XGBoost)** and **C = M5 (Sonnet + classical prior)** unless dev-split evaluation on real data identifies a clear winner among M1 through M6 by calibrated top-1 accuracy on the dev split, locked before test-split unblinding.

8 Calibration and abstention

- Temperature scaling on the dev split.
- Verbalized confidence prompt: model returns a self-rated confidence in {low, medium, high}, mapped to thresholds.
- **Abstention threshold** τ is selected on the dev split as the smallest τ that achieves $\geq 95\%$ precision on covered cases at $\geq 50\%$ coverage. **Locked before test-split unblinding.**

9 Stopping rules

If the dev-split top-1 accuracy of the best LLM-based model is below the XGBoost baseline by more than 3 percentage points, the LLM arm is **not** taken to the clinical study; the paper reports a negative result.

If recruitment yields fewer than 8 dentists, the study is reported as a pilot with descriptive results only; no inferential test is run on H1–H3.

10 Analyses we will NOT run

- We will not split the test set further “to find a subset where C wins.”
- We will not change the abstention threshold τ after seeing the test split.
- We will not switch from exact CDT match to category match after seeing results.
- We will not drop dentists from analysis based on their performance.

Any post-hoc analyses are clearly labeled as exploratory and will not be used to support the primary claims.

11 Reproducibility

- All code open-sourced (MIT) at submission.
- All seeds fixed (`random.seed(42)`; `numpy.seed(42)`; `torch.manual_seed(42)`).
- All LLM calls logged to `data/results_<run>/llm_calls.jsonl` with prompt, response, model, latency, and token counts.
- A reviewer can replay every call from the logs without spending money on the Anthropic API; if they wish to re-execute, they do so through their own Claude Code subscription.
- This pre-registration’s OSF DOI will be cited in the paper’s methods section.

Authors’ declaration. The authors declare that this pre-registration is final at the time of OSF deposit. Any deviations from this plan during data analysis will be documented in the manuscript as exploratory.

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